



Fan Cooling For Improved PERFORMANCE OF REFRIGERATORS

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The refrigerator is one of the most heavily used household appliances, operating continuously throughout the year. Therefore, any improvement in its operating conditions can enhance food cooling performance. Reduced load on the compressor also helps extend its service life, while delivering modest power savings.

Refrigerators remove heat from stored food and dissipate it through the condenser coils. In older models, these coils were mounted on the rear side, where the exposed black coils released heat directly into the surrounding air. In newer versions, the condenser coils are embedded inside both side walls of the refrigerator enclosure. Touching the side walls often reveals warm areas, indicating the location of the coils within the enclosure. Heat from the condenser coils is transferred to these walls, which then radiate it to the outside environment. The wall temperature typically rises to about 10°C above ambient.

A clearance of 5cm to 10cm is recommended on both sides of the refrigerator to allow proper air circulation. However, this clearance is often inadequate due to space constraints. In many cases, the available gap is also used to store household items, which further obstructs airflow. As a result, the refrigerator remains surrounded by warm air, reducing its cooling efficiency.

A simple fan-cooling system addresses this issue. The cooling fans can be magnetically attached to both sides of the fridge using small magnets. The fans switch on automatically whenever the compressor cycle starts. This is achieved by

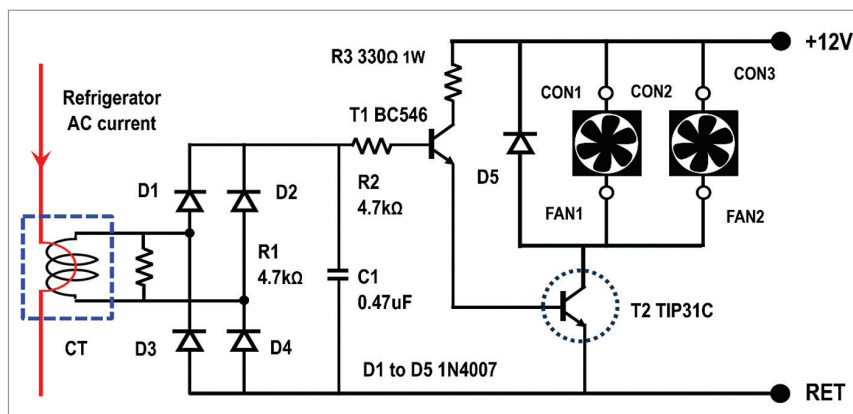


Fig. 1: Circuit diagram of compressor current sensing and fan driver

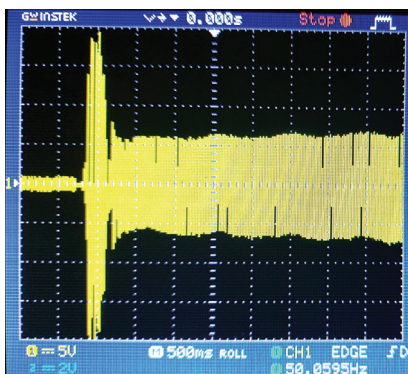


Fig. 2: Starting current waveform of compressor (CT burden 4.7kΩ, primary 2 turns)

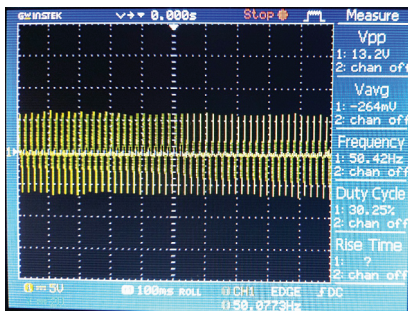


Fig. 3: Running current waveform of the compressor (CT burden 4.7kΩ, primary 2 turns)

sensing the compressor current using a current transformer (CT). When the CT detects current flow, the fans turn on, and they switch off automatically when the compressor cycle ends.

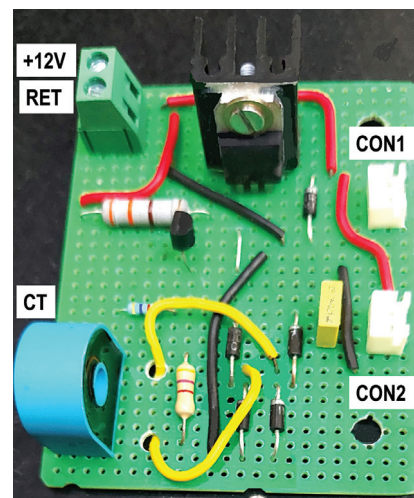


Fig. 4: Controller PCB showing the mounting of CT

Compressor current sensing

The circuit diagram of the proposed system is shown in Fig. 1. It consists of a CT having a ratio of 5A to 5mA output at the secondary. Typically, a compressor draws a few hundred milliamperes. Resistor R1 (4.7kΩ) acts as a burden on the CT. Caution: Do not disconnect the burden resistor from the CT terminals. Without the burden resistor, the CT will generate a high voltage and the coil may burn out.